Notes on Gauge Theory

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1 Affine Connections

1.1 Levi-Civita Connections

When it comes to connections, most of us probably think first of the Levi-Civita connection in Riemannian geometry. So let us first review its most basic definition, and then try to rewrite everything we are familiar with in the language of differential forms.

1.1.1 Covariant Derivative

Definition 1.1. *A connection ∇ on a mannifold M is a map*

$$\begin{aligned}\nabla : C^\infty(M,TM) \times C^\infty(M,TM) &\to C^\infty(M,TM)\\ (Y,X) &\mapsto \nabla_Y X\end{aligned}$$

satisfying the following properties:

$$\begin{aligned}\nabla_{f_1Y_1+f_2Y_2}X =& f_1\nabla_{Y_1}X + f_2\nabla_{Y_2}X,\\ \nabla_Y(\lambda_1X_1+\lambda_2X_2) =& \lambda_1\nabla_Y X_1 + \lambda_2\nabla_Y X_2,\\ \nabla_Y fX =& \mathrm{d}f(Y)\cdot X + f\nabla_Y X.\end{aligned}$$

where $X, X_1, X_2, Y, Y_1, Y_2 \in \Gamma(TM)$, $f, f_1, f_2 \in C^\infty(M)$, and $\lambda_1, \lambda_2 \in \mathbb{R}$.

Definition 1.2. *The **torsion**[1] of a connection ∇ is defined as the tensor*

$$T(X,Y) := \nabla_X Y - \nabla_Y X - [X,Y].$$

*The **curvature** of a connection ∇ is defined as the tensor*

$$R(X,Y)Z := \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z.$$

Definition 1.3. *A connection ∇ of a Riemannian manifold (M,g) is called a Levi-Civita connection if it is torsion-free and compatible with the metric g, i.e.*

$$\begin{aligned}\nabla_X Y - \nabla_Y X = [X,Y],\\ X(g(Y,Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z).\end{aligned}$$

[1] It can only be defined for connections on the tangent bundle, since it involves both $\nabla_X Y$ and $\nabla_Y X$.

1.1.2 Exterior convariant derivative

A connection ∇ can also be viewed as a map

$$\begin{aligned}\nabla : C^\infty(M, TM) &\to C^\infty(M, T^*M \otimes TM)\\ X &\mapsto \nabla X\end{aligned}$$

where ∇X is a 1-form taken value in TM[1] and $(\nabla X)(Y) := \nabla_Y X$. ∇ is not a $C^\infty(M)$-linear map, but

$$\nabla(fX) = \mathrm{d}f \otimes X + f\nabla X.$$

We can extend ∇ to the exterior covariant derivative

$$\begin{aligned}\nabla : \Omega^*(M, TM) &\to \Omega^{*+1}(M, TM)^2\\ \eta &\mapsto \nabla\eta,\end{aligned}$$

satisfying the Leibniz rule:

$$\nabla(\alpha \wedge \eta) = \mathrm{d}\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla\eta, \quad \forall \alpha \in \Omega^q(M), \eta \in \Omega^*(M, TM),$$

For example, let $\eta = \alpha \otimes X$ be a section of $\Omega^r(M, TM)$, then

$$\nabla\eta = \nabla(\alpha \otimes X) = \mathrm{d}\alpha \otimes X + (-1)^r \alpha \wedge \nabla X.$$

Under the above setup, we can define $\nabla^2 = \nabla \circ \nabla, \nabla^3, \ldots, \nabla^k$, and we can prove the following property:

Proposition 1.4. *$\nabla^2 : \Omega^*(M, TM) \to \Omega^{*+2}(M, TM)$ is an $\Omega^*(M)$-linear map, i.e.*

$$\nabla^2(\alpha \wedge \eta) = \alpha \wedge \nabla^2\eta, \qquad \forall \alpha \in \Omega^*(M), \eta \in \Omega^*(M, TM).$$

Thus $R := \nabla^2$ defines a section of $\Omega^2(M, \mathrm{End}(TM))$, which is called the ***curvature*** *of the connection ∇.*

Proof.

$$\begin{aligned}\nabla^2\big(\alpha \wedge \eta\big) =& \nabla\big(\mathrm{d}\alpha \wedge \eta + (-1)^q \alpha \wedge \mathrm{d}^\nabla \eta\big)\\ =& \mathrm{d}^2\alpha \wedge \eta + (-1)^{q+1}\mathrm{d}\alpha \wedge \mathrm{d}^\nabla\eta + (-1)^q \mathrm{d}\alpha \wedge \nabla\eta + \alpha \wedge \nabla^2\eta\\ =& \alpha \wedge \nabla^2\eta.\end{aligned}$$

$\square$

[1] One can simply view ∇X as a $(1,1)$-tensor field.

[2] $\Omega^r(M, TM) := C^\infty(M, \wedge^r T^*M \otimes TM)$.

Unlike the usual exterior derivative d, we have $\nabla^2 \neq 0$ in general. The reason why ∇^2 is called the curvature of ∇ is that

Proposition 1.5. *For $X, Y, Z \in C^\infty(M, TM)$, we have*

$$(\nabla^2 Z)(X, Y) = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

Proof. We will prove it after introducing the connection 1-forms. $\square$

1.1.3 Connection 1-form

Locally, we always have a local frame. Let $\{e_i\}$ be a local frame of TM on an open set $U \subset M$, and $\{\theta^i\}$ be the dual frame. then we can write

$$\nabla_{e_i} e_j = \Gamma_{ij}^k e_k,$$

where Γ_{ij}^k are the Christoffel symbols of the connection ∇ w.r.t. $\{e_i\}$. Define

$$\omega_j^k := \Gamma_{ij}^k \theta^i,$$

we call ω_j^k the **connection 1-form** of ∇ w.r.t. $\{e_i\}$. So

$$\nabla e_j = \omega_j^k \otimes e_k.$$

Let the matrix of 1-forms $\omega = (\omega_j^k)$[1], then we can write the above equation in a more compact way:

$$\nabla(e_1, \ldots, e_n) = (e_1, \ldots, e_n)\,\omega = (e_1, \ldots, e_n) \begin{pmatrix} \omega_1^1 & \cdots & \omega_1^n \\ \vdots & & \vdots \\ \omega_n^1 & \cdots & \omega_n^n \end{pmatrix}$$ [2]

We can compute the curvature in terms of the connection 1-form ω:

$$\begin{aligned} \nabla^2(e_1, \ldots, e_n) =& \nabla\big(\nabla(e_1, \ldots, e_n)\big) = \nabla\big((e_1, \ldots, e_n)\,\omega\big) \\ =& \big(\nabla(e_1, \ldots, e_n)\big) \wedge \omega + (e_1, \ldots, e_n)\,\mathrm{d}\omega \\ =& (e_1, \ldots, e_n)\,\omega \wedge \omega + (e_1, \ldots, e_n)\,\mathrm{d}\omega \\ =& (e_1, \ldots, e_n)\,\big(\mathrm{d}\omega + \omega \wedge \omega\big). \end{aligned}$$

[1] The upper index k is the row index, and the lower index j is the column index.

[2] This is an awkward notation. Due to certain notational reasons, we consider arranging the frame as row vectors, which results in differential forms acting on the right of vectors. For ordinary vectors, we are free to place differential forms on the left or right without introducing sign changes. However, it should be noted that for vector-valued differential forms, placing the differential form on the left or right does make a sign difference.

Define

$$\Omega := \mathrm{d}\omega + \omega \wedge \omega, \tag{1}$$

then Ω is a matrix of 2-forms, and we call Ω the **curvature 2-form** of the connection ∇ w.r.t. $\{e_i\}$.

Proposition 1.6. *Let $R(e_i, e_j)e_k = R^l_{ijk}e_l$, then*

$$\Omega^l_k(e_i, e_j) = R^l_{ijk}.$$

Proof. Since $\Omega^l_k = \mathrm{d}\omega^l_k + \omega^l_m \wedge \omega^m_k$,

$$\begin{aligned}
&(\mathrm{d}\omega^l_k)(e_i, e_j)e_l = e_i(\omega^l_k(e_j))e_l - e_j(\omega^l_k(e_i))e_l - \omega^l_k([e_i, e_j])e_l \\
=&\nabla_{e_i}(\omega^l_k(e_j)e_l) - \omega^l_k(e_j)\nabla_{e_i}e_l - \nabla_{e_j}(\omega^l_k(e_i)e_l) + \omega^l_k(e_i)\nabla_{e_j}e_l - \omega^l_k([e_i, e_j])e_l \\
=&\nabla_{e_i}\nabla_{e_j}e_k - \omega^l_k(e_j)\nabla_{e_i}e_l - \nabla_{e_j}\nabla_{e_i}e_k + \omega^l_k(e_i)\nabla_{e_j}e_l - \omega^l_k([e_i, e_j])e_l,
\end{aligned}$$

and

$$\begin{aligned}
(\omega^l_m \wedge \omega^m_k)(e_i, e_j)e_l =&\omega^l_m(e_i)\omega^m_k(e_j)e_l - \omega^l_m(e_j)\omega^m_k(e_i)e_l \\
=&\omega^m_k(e_j)\nabla_{e_i}e_m - \omega^m_k(e_i)\nabla_{e_j}e_m.
\end{aligned}$$

Thus

$$\Omega^l_k(e_i, e_j)e_l = \nabla_{e_i}\nabla_{e_j}e_k - \nabla_{e_j}\nabla_{e_i}e_k - \nabla_{[e_i,e_j]}e_k = R(e_i, e_j)e_k.$$

$\square$

Corollary 1.7. *This proposition immediately implies Prop 1.5 since*

$$(\nabla^2 e_k)(e_i, e_j) = \Omega^l_k(e_i, e_j)e_l = R(e_i, e_j)e_k.$$

The connection 1-form ω is not a tensor, since it depends on the choice of the local frame. Let $\{\tilde{e_j}\}$ be another local frame, and the transition matrix from $\{e_i\}$ to $\{\tilde{e_j}\}$ is $a = (a^j_i)$, i.e.

$$(\tilde{e}_1, \dots, \tilde{e}_n) = (e_1, \dots, e_n)\, a,$$

then

$$\begin{aligned}
(\tilde{e}_1, \dots, \tilde{e}_n)\tilde{\omega} =&\nabla(\tilde{e}_1, \dots, \tilde{e}_n) = \nabla\big((e_1, \dots, e_n)\, a\big) \\
=&\big(\nabla(e_1, \dots, e_n)\big)a + (e_1, \dots, e_n)\mathrm{d}a \\
=&(e_1, \dots, e_n)\omega a + (e_1, \dots, e_n)\mathrm{d}a \\
=&(\tilde{e}_1, \dots, \tilde{e}_n)(a^{-1}\omega a + \mathrm{d}a).
\end{aligned}$$

Thus

$$\tilde{\omega} = a^{-1}\mathrm{d}a + a^{-1}\omega a. \tag{2}$$

Applying the transformation formula (2) to (1), we get

$$\begin{aligned}\tilde{\Omega} =& \mathrm{d}\tilde{\omega} + \tilde{\omega}\wedge\tilde{\omega}\\ =&\mathrm{d}(a^{-1}\mathrm{d}a + a^{-1}\omega a) + (a^{-1}\mathrm{d}a + a^{-1}\omega a)\wedge(a^{-1}\mathrm{d}a + a^{-1}\omega a)\\ =&\mathrm{d}a^{-1}\wedge\mathrm{d}a + \mathrm{d}a^{-1}\wedge\omega a + a^{-1}\mathrm{d}\omega\, a - a^{-1}\omega\wedge\mathrm{d}a\\ &+ a^{-1}\mathrm{d}a\wedge a^{-1}\mathrm{d}a + a^{-1}\mathrm{d}a\wedge a^{-1}\omega a + a^{-1}\omega\wedge\mathrm{d}a + a^{-1}\omega\wedge\omega a,\end{aligned}$$

since

$$a^{-1}a = I \implies \mathrm{d}a^{-1}a + a^{-1}\mathrm{d}a = 0 \implies \mathrm{d}a^{-1} = -a^{-1}\mathrm{d}a\, a^{-1},$$

we have

$$\begin{aligned}\tilde{\Omega} =& - a^{-1}\mathrm{d}a\, a^{-1}\wedge\mathrm{d}a - a^{-1}\mathrm{d}a\, a^{-1}\wedge\omega a + a^{-1}\mathrm{d}\omega\, a - a^{-1}\omega\wedge\mathrm{d}a\\ &+ a^{-1}\mathrm{d}a\wedge a^{-1}\mathrm{d}a + a^{-1}\mathrm{d}a\wedge a^{-1}\omega a + a^{-1}\omega\wedge\mathrm{d}a + a^{-1}\omega\wedge\omega a\\ =&a^{-1}(\mathrm{d}\omega + \omega\wedge\omega)a = a^{-1}\Omega a.\end{aligned}$$

Thus transformation formula for curvature between $\{e_i\}$ and $\{\tilde{e}_j\}$ is given by

$$\tilde{\Omega} = a^{-1}\Omega a. \tag{3}$$

The above formula shows that although Ω isn't a globally defined form, two different curvature forms only differ by a conjugation. So if we apply a quantity invariant under conjugation, we obtain a globally defined form. This is the idea of Chern-Weil theory, which we will discuss in later section.

For a Riemannian manifold (M, g), let ∇ be the Levi-Civita connection. We can take $\{e_i\}$ to be a local orthonormal frame, and $\{\omega^i\}$ be the corresponding dual frame. Then

$$\begin{aligned}&\langle\nabla_{e_k}e_i, e_j\rangle + \langle e_i, \nabla_{e_k}e_j\rangle = e_k\langle e_i, e_j\rangle = 0\\ \implies&\langle\omega_i^l(e_k)e_l, e_j\rangle + \langle e_i, \omega_j^l(e_k)e_l\rangle = 0\\ \implies&\omega_i^j(e_k) + \omega_j^i(e_k) = 0.\end{aligned}$$

So the connection 1-form ω w.r.t. an orthonormal frame is skew-symmetric, i.e. $\omega\in\Omega^1(M, \mathfrak{so}(n))$. Since $\Omega = \mathrm{d}\omega + \omega\wedge\omega$, the curvature 2-form Ω is also skew-symmetric, i.e. $\Omega\in\Omega^2(M, \mathfrak{so}(n))$.

1.2 General Affine Connections

Let $\pi : E \to M$ be a rank m vector bundle, we can also define affine connections on E in a similar way as we did for the tangent bundle.

1.2.1 Covariant Derivative

The content of this section is almost a verbatim copy of the previous content.

Definition 1.8. *An (affine) connection ∇ on a vector bundle E is a map*

$$\begin{aligned} \nabla : C^\infty(M, TM) \times C^\infty(M, E) &\to C^\infty(M, E) \\ (X, s) &\mapsto \nabla_X s \end{aligned}$$

satisfying

$$\begin{aligned} \nabla_{f_1 X_1 + f_2 X_2} s =& f_1 \nabla_{X_1} s + f_2 \nabla_{X_2} s, \\ \nabla_X(\lambda_1 s_1 + \lambda_2 s_2) =& \lambda_1 \nabla_X s_1 + \lambda_2 \nabla_X s_2, \\ \nabla_X(fs) =& \mathrm{d}f(X) \cdot s + f \nabla_X s, \end{aligned}$$

The existence of connections on any vector bundle can be proved by using partitions of unity, which we left as an exercise.

The reason why we call ∇ an **affine** connection is that given two connections ∇_1 and ∇_2 on E, $(1-t)\nabla_1 + t\nabla_2$ is also a connection on E for any $t \in \mathbb{R}$.

Let $(\nabla s)(X) := \nabla_X s$, we can extend ∇ to the exterior covariant derivative

$$\begin{aligned} \nabla : \Omega^*(M, E) &\to \Omega^{*+1}(M, E) \\ \eta &\mapsto \nabla \eta, \end{aligned}$$

satisfying the Leibniz rule:

$$\nabla(\alpha \wedge \eta) = \mathrm{d}\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla \eta, \quad \forall \alpha \in \Omega^q(M), \eta \in \Omega^*(M, E).$$

Proposition 1.9. *$\nabla^2 : \Omega^*(M, E) \to \Omega^{*+2}(M, E)$ is an $\Omega^*(M)$-linear map, i.e.*

$$\nabla^2(\alpha \wedge \eta) = \alpha \wedge \nabla^2 \eta, \qquad \forall \alpha \in \Omega^*(M), \eta \in \Omega^*(M, E).$$

Proof. For $\alpha \in \Omega^q(M), \eta \in \Omega^*(M, E)$,

$$\begin{aligned} \nabla^2\big(\alpha \wedge \eta) =& \nabla(\mathrm{d}\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla \eta\big) \\ =& \mathrm{d}^2\alpha \wedge \eta + (-1)^{q+1} \mathrm{d}\alpha \wedge \nabla \eta + (-1)^q \mathrm{d}\alpha \wedge \nabla \eta + \alpha \wedge \nabla^2 \eta \\ =& \alpha \wedge \nabla^2 \eta. \end{aligned}$$

□

Definition 1.10. *The curvature of a connection ∇ on E is defined as $R := \nabla^2 \in \Omega^2(M, \mathrm{End}(E))$.*

1.2.2 Connection 1-form

Let $\{e_i\}$ be a local frame of E on an open set $U \subset M$. Define ω_j^k by

$$\nabla_X e_j = \omega_j^k(X) e_k,$$

this 1-form matrix $\omega = (\omega_j^k)$ is called **the connection 1-form** of ∇ w.r.t. $\{e_i\}$. Then we can write the above equation in a more compact way:

$$\nabla(e_1, \dots, e_m) = (e_1, \dots, e_m)\, \omega = (e_1, \dots, e_m) \begin{pmatrix} \omega_1^1 & \cdots & \omega_1^m \\ \vdots & & \vdots \\ \omega_m^1 & \cdots & \omega_m^m \end{pmatrix}.$$

1.2.3 Horizontal Distribution

In this subsection, we give another interpretation of a connection on a vector bundle, namely a horizontal distribution, and prove that it is equivalent to the definition given in terms of covariant derivatives.

Let $\pi : E \to M$ be a rank m vector bundle, and $p \in E_x$ be a point in the fibre over $x \in M$. The propjection π induces a linear map $\pi_{*p} : T_pE \to T_xM$.

Proposition 1.11. *The kernel of π_* form a subbundle of TE, which is isomorphic to π^*E. We call it the* ***vertical bundle*** *of E.*

Proof. A point $(p, p') \in \pi^*E$ must satisfy $\pi(p) = \pi(p')$, so we can define a curve $\gamma(t) = p + tp'$ in E. Then $\gamma'(0) \in T_pE$ gives a tangent vector at p, this gives an injective morphism from π^*E to TE. So we can view π^*E as a subbundle of TE. Moreover, since $\gamma(t) \subset E_{\pi(p)}$, we have $\pi(\gamma(t)) = \pi(p)$, thus $\pi_{*p}(\gamma'(0)) = 0$. Therefore, $\pi^*E \subset \ker \pi_*$

Since π_{*p} is a surjective linear map, we have

$$\dim \ker \pi_{*p} = \dim T_pE - \dim T_{\pi(p)}M = m = (\pi^*E)_p.$$

Thus $\pi^*E = \ker \pi_*$. $\square$

Let π^*TM be the pull-back of TM by π, then $\pi_* : TE \to TM$ can be viewed as a morphism $\pi_* : TE \to \pi^*TM$. The above proposition implies the following short exact sequence of vector bundles over E:

$$0 \to \pi^*E \to TE \xrightarrow{\pi_*} \pi^*TM \to 0.$$

In smooth category, all short exact sequences of vector bundles split, so we can find a subbundle $H \hookrightarrow TE$ such that

$$TE = H \oplus \pi^*E, \qquad \pi_* : H \to \pi^*TM \text{ is an isomorphism.}$$

Definition 1.12. *A **horizontal distribution** of E is a smooth subbundle $H \subset TE$ satisfying the following properties*

1. $\pi_* : H \to \pi^*TM$ *is an isomorphism, or equivalently, there exists a splitting map* $j : \pi^*TM \to TE$ *such that* $\pi_* \circ j = \mathrm{id}$.

$$0 \longrightarrow \pi^*E \xhookrightarrow{\iota} TE \underset{j}{\overset{\pi_*}{\rightleftarrows}} \pi^*TM \longrightarrow 0$$

2. *H is invariant under the scalar multiplication on E, i.e. for any $\lambda \in \mathbb{R}$. Denote the scalar multiplication by $m_\lambda : E \to E$, $m_\lambda(p) = \lambda p$, then*

$$m_{\lambda * p} H_p = H_{\lambda p}, \quad \forall p \in E.$$

3. *H is invariant under the addition on E, i.e. for any $p, q \in E$ such that $\pi(p) = \pi(q)$, denote the addition by $a_q : E_{\pi(p)} \to E_{\pi(p)}$, $a_q(p) = p + q$, then*

$$a_{q*p} H_p = H_{p+q}, \quad \forall p, q \in E \text{ such that } \pi(p) = \pi(q).$$

The last two conditions are saying that H is compatible with the vector bundle structure of E.

There is no canonical way to choose a horizontal distribution, so a horizontal distribution is an extra structure on E.

Next, we look at how the connection on E gives rise to a horizontal distribution.

Let ∇ be a connection on E. Given any smmoth curve $\gamma : (-\varepsilon, \varepsilon) \to M$ with $\gamma(0) = x$, we say $\tilde{\gamma} : (-\varepsilon, \varepsilon) \to E$ is a **lift** of γ if $\pi(\tilde{\gamma}(t)) = \gamma(t)$ for all t.

Definition 1.13. *A lift $\tilde{\gamma}$ of γ is called a **horizontal lift** if*

$$\nabla_{\gamma'(t)} \tilde{\gamma}(t) = 0, \quad \forall t \in (-\varepsilon, \varepsilon).$$

Proposition 1.14. *For any (picewise) smooth curve $\gamma : (-\varepsilon, \varepsilon) \to M$ and any point $p \in E_{\gamma(0)}$, there exists a unique horizontal lift $\tilde{\gamma}$ of γ starting at p.*

Sketch of proof. The proof is exactly the same as the existence and uniqueness of geodesics. The outline of the proof is to write the equations as a system of ODEs in a local frame; the conclusion then follows from the existence and uniqueness theory of linear ODEs. □

For any vector $v \in T_xM$ and any point $p \in E_x$, we can find a curve γ such that $\gamma(0) = x$ and $\gamma'(0) = v$, the horizontal lift $\tilde{\gamma}$ gives a tangent vector $\tilde{\gamma}'(0) \in T_pE$.

Fact 1. $\tilde{\gamma}'(0) \in T_pE$ is independent of the choice of γ.

Sketch of proof. By solving the ODEs, we know that $\tilde{\gamma}'(0)$ depends only on the derivative of γ at 0, which is $\gamma'(0) = v$. $\square$

So there is a well-defined map

$$j_p : T_xM \to T_pE, \quad j_p(v) := \tilde{\gamma}'(0).$$

By definition,

$$\pi(\tilde{\gamma}(t)) = \gamma(t) \implies \pi_{*p}(j_p(v)) = \pi_{*p}(\tilde{\gamma}'(0)) = \gamma'(0) = v.$$

So j_p is a linear injection from T_xM to T_pE, whose image forms a distribution on E denoted by H.

Fact 2. H is a smooth subbundle of TE.

Sketch of proof. This fact relies on the smooth dependence of solutions of ODEs on initial conditions and parameters. $\square$

Fact 3. H is invariant under the scalar multiplication and addition on E.

Proof. Let $p, q \in E$ such that $\pi(p) = \pi(q)$, $\lambda \in \mathbb{R}$, and γ be a curve such that $\gamma(0) = \pi(p)$. Let $\tilde{\gamma}_p, \tilde{\gamma}_q$ be the horizontal lift of γ starting at p, q respectively, then by the uniqueness of horizontal lift, $\lambda\tilde{\gamma}_p$ is the horizontal lift of γ starting at λp, and $\tilde{\gamma}_p + \tilde{\gamma}_q$ is the horizontal lift of γ starting at $p + q$. Thus

$$m_{\lambda * p}\tilde{\gamma}'(0) = (\lambda\tilde{\gamma}_p)'(0) \in H_{\lambda p},$$
$$a_{q*p}\tilde{\gamma}'(0) = (\tilde{\gamma}_p + \tilde{\gamma}_q)'(0) \in H_{p+q}.$$

$\square$

The above discussion is summarized in the following proposition.

Proposition 1.15. *H is a horizontal distribution of E.*

Conversaly, given a horizontal distribution H, we can derive a connection ∇ on E as follows.

Definition 1.16. *For any $p \in E$ and $v \in TE$, we can write $v = v_H + v_V$ where $v_H \in H$ and $v_V \in \pi^*E$. We call v_H and v_V the horizontal and vertical component of v respectively.*

For any $X \in C^\infty(M, TM)$ and $s \in C^\infty(M, E)$, the tangent map of s gives a map $s_* : TM \to TE$, then we can define $\nabla_X s$ by

$$\nabla_X s := s_*(X)_V = \text{the vertical component of } s_*(X).$$

Exercises

Exercise 1.1. In Riemannian geometry, we can define the covariant derivative of a 1-form ω by

$$(\nabla_X \omega)(Y) := X(\omega(Y)) - \omega(\nabla_X Y).$$

For an (r, s)-tensor field T, the covariant derivative $\nabla_Z T$ is

$$\begin{aligned}(\nabla_Z T)(X_1, \dots, X_r, \omega^1, \dots, \omega^s) : = \; & Z(T(X_1, \dots, X_r, \omega^1, \dots, \omega^s)) \\ & - \sum_{i=1}^{r} T(X_1, \dots, \nabla_Z X_i, \dots, X_r, \omega^1, \dots, \omega^s) \\ & - \sum_{j=1}^{s} T(X_1, \dots, X_r, \omega^1, \dots, \nabla_Z \omega^j, \dots, \omega^s).\end{aligned}$$

If we rearrange the terms, we see that this is precisely the Leibniz rule.

The curvature operator can also act on general tensors

$$R(X, Y)T := \nabla_X \nabla_Y T - \nabla_Y \nabla_X T - \nabla_{[X,Y]} T.$$

We define the covariant differential $\nabla_{\text{tensor}} T$ as the $(r, s+1)$-tensor field

$$(\nabla_{\text{tensor}} T)(Z, \cdots, X_i, \cdots, \cdots, \omega^j, \cdots) := (\nabla_Z T)(\cdots, X_i, \cdots, \cdots, \omega^j, \cdots)$$

This is another way to extend the covariant derivative to tensor fields, and we denote it by ∇_{tensor} to distinguish it from the exterior covariant derivative.

Prove that for a torsion-free connection, ∇_{tensor} satisfies the Ricci identity

$$(\nabla^2_{\text{tensor}} T)(X, Y, \cdots, \cdots) - (\nabla^2_{\text{tensor}} T)(Y, X, \cdots, \cdots) = (R(X, Y)T)(\cdots, \cdots)$$

Exercise 1.2. For a vector field $Z \in C^\infty(M, TM)$, both $\nabla^2 Z$ and $\nabla^2_{\text{tensor}} Z$ are $(1, 2)$-tensor fields, but by Ricci identity, we have

$$(\nabla^2_{\text{tensor}} Z)(X, Y) - (\nabla^2_{\text{tensor}} Z)(Y, X) = R(X, Y)Z = (\nabla^2 Z)(X, Y).$$

Can you explain this phenomenon?

Exercise 1.3 (Torsion 2-form)**.** Let ∇ be a connection on TM, $\{e_i\}$ be a local frame of TM, $\{\theta^i\}$ be the corresponding dual coframe and ω be the connection 1-form of ∇ w.r.t. $\{e_i\}$. Define

$$\tau^i := \mathrm{d}\theta^i + \omega^i_j \wedge \theta^j.$$

If we arrange $\theta = (\theta^1, \dots, \theta^n)^T$ as a column vector, then the above formula can be written as

$$\tau = \mathrm{d}\theta + \omega \wedge \theta.$$

then τ is called the **torsion 2-form** of ∇ w.r.t. $\{e_i\}$.

Let $T(e_i, e_j) = T_{ij}^k e_k$, where T is the torsion of ∇. Prove that

$$\tau^k(e_i, e_j) = T_{ij}^k.$$

Thus

$$\tau^k = \frac{1}{2} T_{ij}^k \theta^i \wedge \theta^j.$$

Exercise 1.4 (Levi-Civita connection)**.** Let (M, g) be a Riemannian manifold, ∇ be a connection on TM. Show that

1. ∇ is torsion-free if and only if $\tau = 0$, i.e.,

$$\mathrm{d}\theta^i + \omega_j^i \wedge \theta^j = 0.$$

2. If $\{e_i\}$ is a local orthonormal frame, then ∇ is compatible with g if and only if ω is skew-symmetric, i.e.,

$$\omega_j^i + \omega_i^j = 0.$$

3. Under a local orthonormal frame, there exists a unique connection 1-form ω satisfying

$$\mathrm{d}\theta^i + \omega_j^i \wedge \theta^j = 0, \quad \omega_j^i + \omega_i^j = 0.$$

This is the existence and uniqueness of the Levi-Civita connection.

Exercise 1.5. Let ∇ be a connection on TM, $\{e_i\}$ be a local frame of TM, $\{\theta^i\}$ be the corresponding dual coframe and ω be the connection 1-form of ∇ w.r.t. $\{e_i\}$. The following two equations

$$\begin{aligned} \mathrm{d}\theta =& \tau - \omega \wedge \theta, \\ \mathrm{d}\omega =& \Omega - \omega \wedge \omega, \end{aligned}$$

are called the **structure equations**. Show that τ and Ω satisfy

$$\begin{aligned} \mathrm{d}\tau =& \Omega \wedge \theta - \omega \wedge \tau, \\ \mathrm{d}\Omega =& \Omega \wedge \omega - \omega \wedge \Omega. \end{aligned}$$

These two equations are called the **Bianchi identities**. Furthermore, show that

$$\mathrm{d}(\Omega^k) = \Omega^k \wedge \omega - \omega \wedge \Omega^k.$$

Exercise 1.6. Show that if ∇ is torsion-free, then $\mathrm{d}\tau = \Omega \wedge \theta - \omega \wedge \tau$ is equivalent to the first Bianchi identity

$$R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = 0.$$

Exercise 1.7. Show that if ∇ is torsion-free, then $\mathrm{d}\Omega = \Omega \wedge \omega - \omega \wedge \Omega$ is equivalent to the second Bianchi identity

$$(\nabla_X R)(Y,Z) + (\nabla_Y R)(Z,X) + (\nabla_Z R)(X,Y) = 0.$$

Exercise 1.8. For $\{e_i\} = \{\frac{\partial}{\partial x^i}\}$, $\{\tilde{e}_j\} = \{\frac{\partial}{\partial \tilde{x}^j}\}$, $\omega_j^k = \Gamma_{ij}^k \mathrm{d}x^i$, show that the transformation formula

$$\tilde{\omega} = a^{-1}\mathrm{d}a + a^{-1}\omega a$$

reduces to the well-known transformation formula of Christoffel symbols:

$$\tilde{\Gamma}_{ij}^k = \frac{\partial \tilde{x}^k}{\partial x^l} \frac{\partial x^m}{\partial \tilde{x}^i} \frac{\partial x^n}{\partial \tilde{x}^j} \Gamma_{mn}^l + \frac{\partial^2 x^l}{\partial \tilde{x}^i \partial \tilde{x}^j} \frac{\partial \tilde{x}^k}{\partial x^l}.$$